Transient and Steady-State – Worked Problems

Practice: Routh stability ranges and transient specifications.

Ogata, *Modern Control Engineering*, Ch. 5 (end-of-chapter style).

Step through with **Ctrl+Enter**, or render a report with `publish`.

Contents

Problem 1: Routh Stability Range

A unity-feedback system has forward-path transfer function

$G(s) = \frac{K}{s(s+2)(s+4)}$

Find the range of $K > 0$ for closed-loop stability.

Characteristic equation: $s(s+2)(s+4) + K = 0 \Rightarrow s^3 + 6s^2 + 8s + K = 0$

Routh array:

```
s\^{}3 \ensuremath{|}  1     8
s\^{}2 \ensuremath{|}  6     K
s\^{}1 \ensuremath{|}  (48-K)/6
s\^{}0 \ensuremath{|}  K
```

Requiring all first-column entries positive: $K > 0$ and $48 - K > 0 \Rightarrow K < 48$. So $0 < K < 48$.

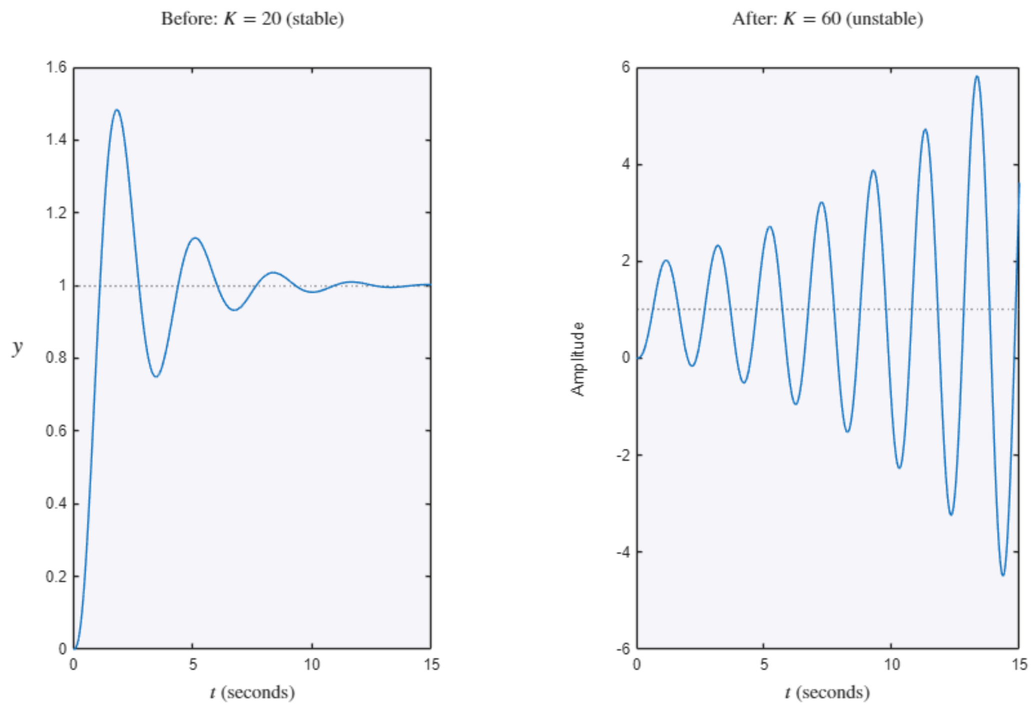
```
syms s K
charpoly1 = expand(s*(s+2)*(s+4)+K)

Ks = [10 30 47.9 48 48.1];
for k = Ks
    p = roots([1 6 8 k]);
    fprintf('K = %5.1f -> max real part = %8.4f\n', k, max(real(p)))
end
```

```
charpoly1 =

s^3 + 6*s^2 + 8*s + K

K =  10.0 -> max real part =  -0.6196
K =  30.0 -> max real part =  -0.2330
K =  47.9 -> max real part =  -0.0011
K =  48.0 -> max real part =   0.0000
K =  48.1 -> max real part =   0.0011
```

Problem 1 – Before vs. After: Stable vs. Unstable Gain

In the time domain: K=20 (< 48) settles; K=60 (> 48) diverges – the Routh range made visible.

```
t_cl = 0:0.01:15;
figure
subplot(1,2,1)
step(feedback(tf(20,[1 6 8 0]),1), t_cl)
title('Before: $K=20$ (stable)','Interpreter','latex','FontSize',13)
ylabel('$y$','Interpreter','latex','FontSize',15)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$','Interpreter','latex','FontSize',13)
subplot(1,2,2)
step(feedback(tf(60,[1 6 8 0]),1), t_cl)
title('After: $K=60$ (unstable)','Interpreter','latex','FontSize',13)
xlabel('$t$','Interpreter','latex','FontSize',13)
```

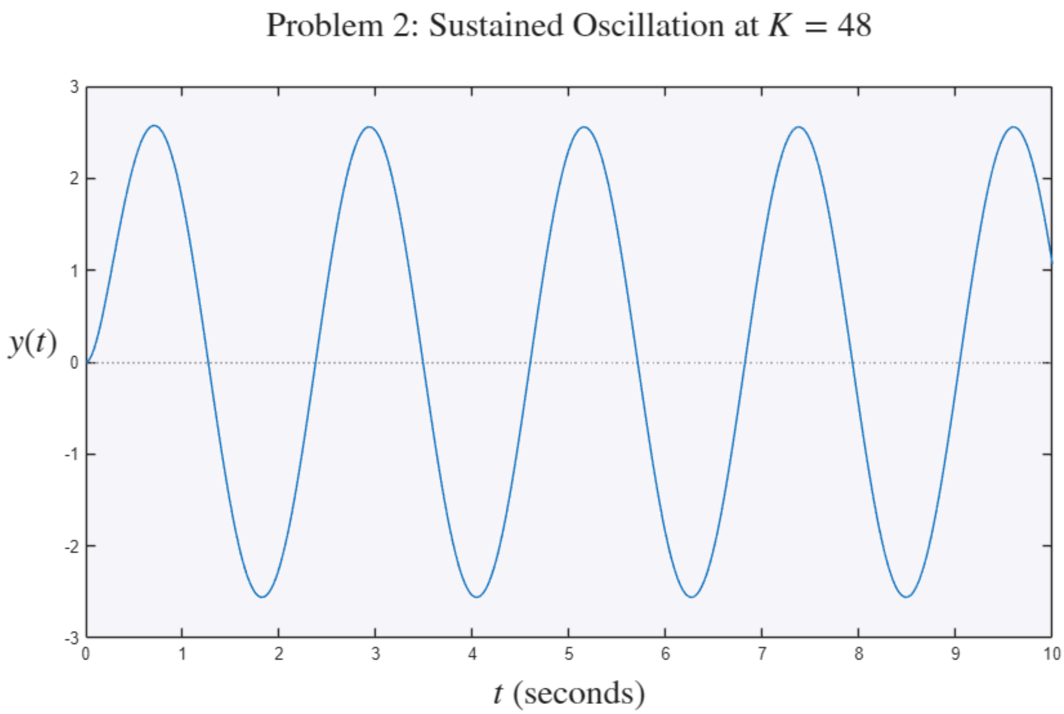


Problem 2: Marginal Stability and Sustained Oscillation

Using the boundary $K = 48$ from Problem 1, the system is marginally stable; the row of zeros at s^1 indicates a pair of poles on the $j\omega$-axis. The auxiliary polynomial is formed from the s^2 row: $6s^2 + K = 0 \Rightarrow s^2 = -K/6 \Rightarrow s = \pm j\sqrt{K/6}$.

```
K_marginal = 48;
p_marginal = roots([1 6 8 K_marginal]);
disp('Closed-loop poles at K = 48:'); disp(p_marginal)
omega_pred = sqrt(K_marginal/6);
fprintf('Auxiliary-polynomial predicted omega = %.4f rad/s\n', omega_pred)

G_marginal = tf(K_marginal,[1 6 8 0]);
T_marginal = feedback(G_marginal,1);
figure
impulse(T_marginal, 0:0.01:10)
title('Problem 2: Sustained Oscillation at $K=48$','Interpreter','latex','FontSize',20)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$','Interpreter','latex','FontSize',20)
```

```
Closed-loop poles at K = 48:
  -6.0000 + 0.0000i
   0.0000 + 2.8284i
   0.0000 - 2.8284i

Auxiliary-polynomial predicted omega = 2.8284 rad/s
```

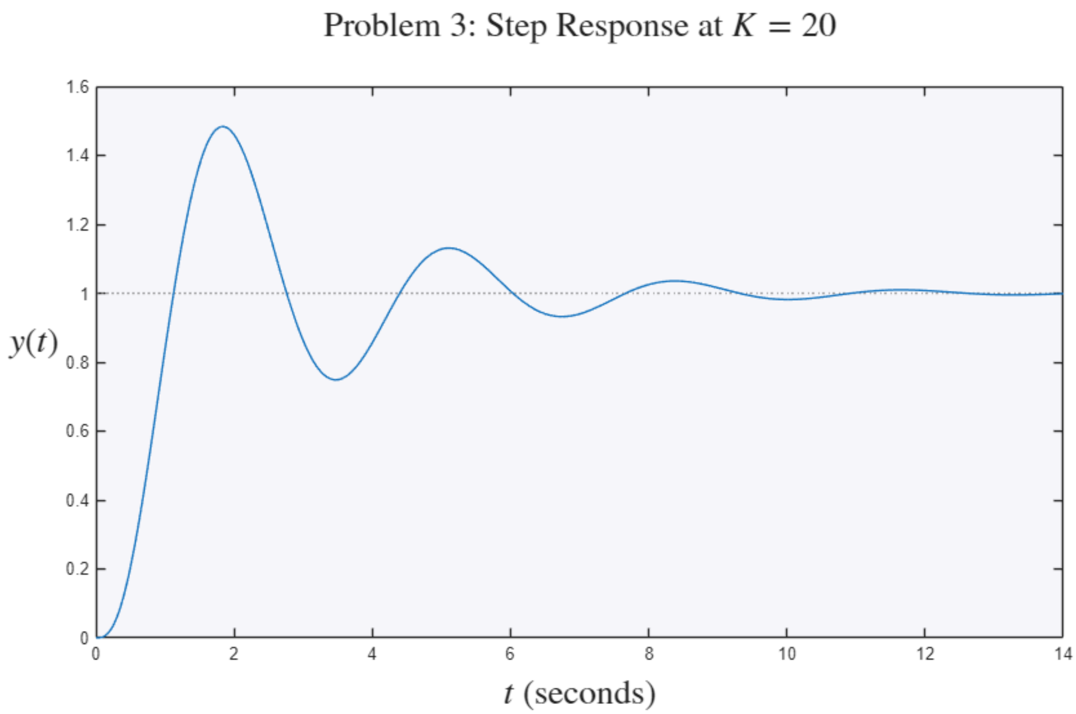


Problem 3: Transient Specs from a Gain Choice

For the same plant family, choose $K = 20$ (well inside the stable range found in Problem 1) and find the closed-loop step-response specifications.

```
K3 = 20;
G3 = tf(K3,[1 6 8 0]);
T3 = feedback(G3,1);
poles_3 = pole(T3)

info3 = stepinfo(T3)

figure
step(T3)
title('Problem 3: Step Response at $K=20$','Interpreter','latex','FontSize',20)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$','Interpreter','latex','FontSize',20)
```

```
poles_3 =

  -5.2012 + 0.0000i
  -0.3994 + 1.9198i
  -0.3994 - 1.9198i


info3 =

  struct with fields:

         RiseTime: 0.6760
    TransientTime: 8.9125
     SettlingTime: 8.9125
      SettlingMin: 0.7486
      SettlingMax: 1.4833
        Overshoot: 48.3323
       Undershoot: 0
             Peak: 1.4833
         PeakTime: 1.8416
```



Try it yourself

- In Problem 1, raise the gain and watch the max-real-part curve cross zero exactly at the Routh limit K=48.
- In Problem 3, try K=40 (closer to the boundary) and notice the extra overshoot and slower settling reported by `stepinfo`.